

Name _____ Surname _____ Date ____/____/____

Exam-style questions

Exercises 1 and 2 are worked together on the board, write your answers here as we go.

Exercise 1

Rolling ball

A student times a ball rolling down a ramp five times. Her results are 1.84 s, 1.79 s, 2.31 s, 1.82 s and 1.80 s.

(a) Identify the anomalous result. [1]

..... s

(b) Calculate the mean time, ignoring the anomalous result. Give your answer to 3 significant figures. [2]

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mean / media = s

(c) Suggest one reason why her results are not all the same. [1]

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Exercise 2

The balance that isn't zero

La bilancia che non parte da zero

With nothing on it, a digital balance reads 0.3 g. A student places a stone on the balance and it reads 25.6 g.

(a) State the name of this type of error. [1]

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(b) Calculate the correct mass of the stone. [1]

..... g

(c) The student weighs the stone five times and takes the mean. Explain whether this removes the error. [1]

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Exercise 3

A thermometer has a scale marked every 1 °C.

(a) State the smallest change in temperature that can be read reliably from this thermometer. [1]

..... °C

(b) The top of the liquid is halfway between the 23 °C and 24 °C marks. Record the temperature. [1]

..... °C

Exercise 4

Six students each time the same single run of a toy car over 1.00 m. Their results are: 2.46 s, 2.52 s, 2.49 s, 2.95 s, 2.43 s, 2.50 s.

(a) Identify the anomalous result. [1]

..... s

(b) Calculate the mean time, ignoring the anomalous result. [2]

.....
mean / media = s

(c) Calculate the range of the results, ignoring the anomalous result. [1]

range = s

(d) Suggest why the teacher asked all six students to time the same run, instead of six different runs. [1]

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(e) Suggest one change to the method that would reduce the spread of the results. [1]

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Exercise 5

For each situation, state whether it causes a random error or a systematic error. Give a reason for each answer. [4]

(a) Each time, a student starts the stopwatch slightly early or slightly late.

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(b) The end of a ruler is worn away, so its edge is at the 2 mm mark. A student always lines objects up with the edge.

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(c) A student always reads a measuring cylinder with her eye above the level of the liquid.

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(d) Gusts of air sometimes speed up and sometimes slow down a falling paper cone.

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Exercise 6

The true time of a swing is 2.50 s. A student's reactions are not perfect.

(a) He starts the stopwatch 0.20 s late and stops it exactly on time. State whether his reading is too long or too short, and by how much. [1]

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(b) Next time he starts it 0.20 s late and stops it 0.20 s late. What does the stopwatch read? [1]

..... s

(c) Then he starts it 0.20 s late and stops it 0.30 s late. Calculate the reading. [1]

..... s

(d) Explain why reaction times that change from one reading to the next produce a random error. [1]

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